Requirements Engineering

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Requirements Engineering Task

- Inception—ask a set of questions that establish …
  - basic understanding of the problem
  - the people who want a solution
  - the nature of the solution that is desired, and
  - the effectiveness of preliminary communication and collaboration between the customer and the developer
- Elicitation—elicit requirements from all stakeholders
- Elaboration—create an analysis model that identifies data, function and behavioral requirements
- Negotiation—agree on a deliverable system that is realistic for developers and customers

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Requirements Engineering Task

- Specification—can be any one (or more) of the following:
  - A written document
  - A set of models
  - A formal mathematical
  - A collection of user scenarios (use-cases)
  - A prototype
- Validation—a review mechanism that looks for
  - errors in content or interpretation
  - areas where clarification may be required
  - missing information
  - inconsistencies (a major problem when large products or systems are engineered)
  - conflicting or unrealistic (unachievable) requirements.

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Requirements Engineering Task

- Requirements management

  A set of activities that help the project team identify, control, and track requirements and changes to requirements at any time as the project proceeds.

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Inception

- Identify stakeholders
  - "who else do you think I should talk to?"
- Recognize multiple points of view
- Work toward collaboration
- The first questions
  - Who is behind the request for this work?
  - Who will use the solution?
  - What will be the economic benefit of a successful solution
  - Is there another source for the solution that you need?

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Eliciting Requirements

Why requirements elicitation is difficult ?

- Problems of scope.
- Problems of understanding
- Problems of volatility

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Eliciting Requirements

- meetings are conducted and attended by both software engineers and customers
- rules for preparation and participation are established
- an agenda is suggested
- a "facilitator" (can be a customer, a developer, or an outsider) controls the meeting
- a "definition mechanism" (can be work sheets, flip charts, or wall stickers or an electronic bulletin board, chat room or virtual forum) is used
- the goal is
  - to identify the problem
  - propose elements of the solution
  - negotiate different approaches, and
  - specify a preliminary set of solution requirements

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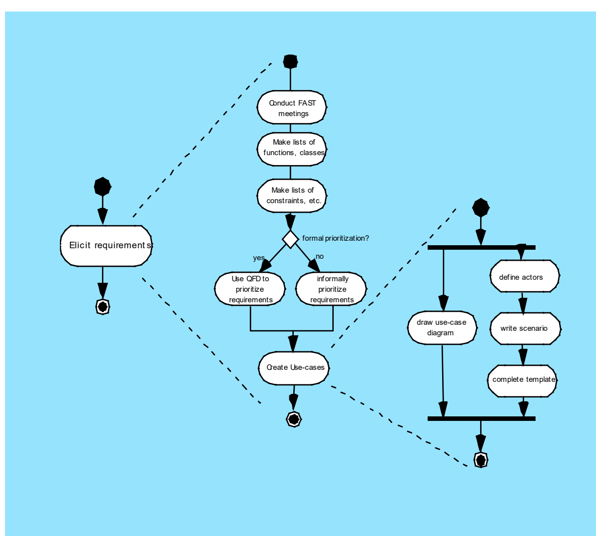
Eliciting Requirements



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Quality Function Deployment

- Normal Requirement: These requirements reflect objectives and goals stated for a product or system during meetings with customer. If the requirements are present, the customer is satisfied.
- Expected Requirement: These requirements are implicit to the product or system. The customer does not explicitly state them. There absence will be cause for significant dissatisfaction.

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Quality Function Deployment

- Exciting requirements: These requirements reflect features that go beyond the customer's expectations and prove to be very satisfying when present.

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Elicitation Work Products

- a statement of need and feasibility.
- a bounded statement of scope for the system or product.
- a list of customers, users, and other stakeholders who participated in requirements elicitation
- a description of the system's technical environment.
- a list of requirements (preferably organized by function) and the domain constraints that apply to each.
- a set of usage scenarios that provide insight into the use of the system or product under different operating conditions.
- any prototypes developed to better define requirements.

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Use-Cases

- A collection of user scenarios that describe the thread of usage of a system
- Each scenario is described from the point-of-view of an "actor"—a person or device that interacts with the software in some way
- Each scenario answers the following questions:
  - Who is the primary actor, the secondary actor (s)?
  - What are the actor's goals?
  - What preconditions should exist before the story begins?
  - What main tasks or functions are performed by the actor?
  - What extensions might be considered as the story is described?
  - What variations in the actor's interaction are possible?
  - What system information will the actor acquire, produce, or change?
  - Will the actor have to inform the system about changes in the external environment?
  - What information does the actor desire from the system?
  - Does the actor wish to be informed about unexpected changes?

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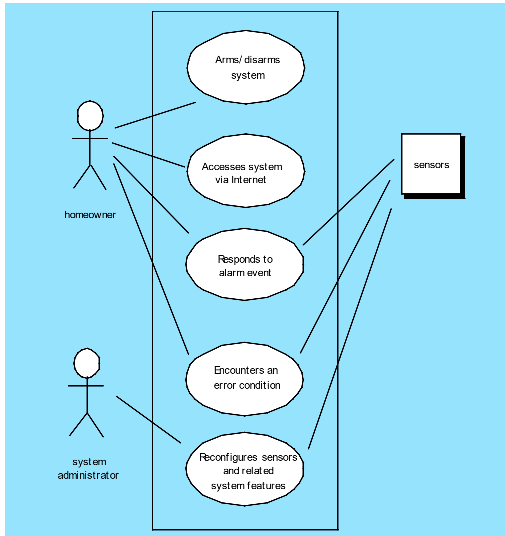
Use-Case Diagram



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Elaboration

- Elaboration is an analysis modeling action
- Refine user scenarios
- User scenarios is parsed to extract analysis model

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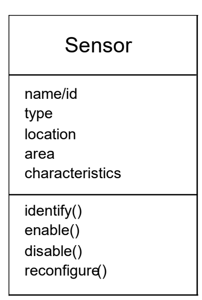
Building the Analysis Model

- Elements of the analysis model
  - Scenario-based elements
    - Functional—processing narratives for software functions
    - Use-case—descriptions of the interaction between an “actor” and the system
  - Class-based elements
    - Implied by scenarios
  - Behavioral elements
    - State diagram
  - Flow-oriented elements
    - Data flow diagram

Class Diagram

From the *SafeHome* system …



State Diagram

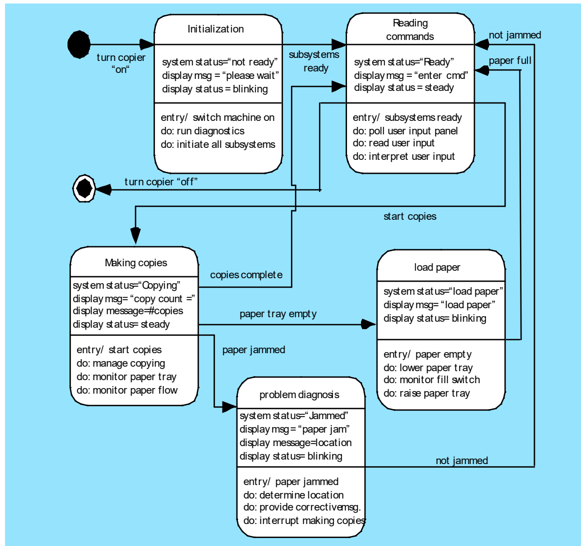


Figure 7.6 Preliminary UML state diagram for a photocopier

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Negotiating Requirements

- Identify the key stakeholders
  – These are the people who will be involved in the negotiation
- Determine each of the stakeholders "win conditions"
  – Win conditions are not always obvious
- Negotiate
  – Work toward a set of requirements that lead to "win-win"

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Validating Requirements-I

- Is each requirement consistent with the overall objective for the system/product?
- Have all requirements been specified at the proper level of abstraction? That is, do some requirements provide a level of technical detail that is inappropriate at this stage?
- Is the requirement really necessary or does it represent an add-on feature that may not be essential to the objective of the system?
- Is each requirement bounded and unambiguous?
- Does each requirement have attribution? That is, is a source (generally, a specific individual) noted for each requirement?
- Do any requirements conflict with other requirements?

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Validating Requirements-II

- Is each requirement achievable in the technical environment that will house the system or product?
- Is each requirement testable, once implemented?
- Does the requirements model properly reflect the information, function and behavior of the system to be built.
- Has the requirements model been "partitioned" in a way that exposes progressively more detailed information about the system.
- Have requirements patterns been used to simplify the requirements model. Have all patterns been properly validated? Are all patterns consistent with customer requirements?

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Requirements Management

- Identification: each requirement is assigned a unique identifier
- Create traceability table
  – Feature traceability table: show how requirements relate to important customer observable/product features
  – Source traceability table: Identifies the source of each requirement
  – Dependency traceability table: Indicates how requirements are related to one another
  – Subsystem traceability table: Categorizes requirements by the subsystem(s) that they govern

Requirements Management

  – Interface traceability table: Shows how requirements relate to both internal and external system interfaces
- Traceability tables are used to understand how a change in one requirement will affect different aspects of the system to be built